Modelling Simulation Optimisation for Delivery Driver

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***Abstract*—The demand for delivery drivers has greatly expanded as a result of the rapid expansion of e-commerce and online food delivery platforms. However, this line of work comes with its own set of difficulties, including rushed timelines, erratic traffic, and variable delivery routes. The purpose of this study is to boost operational efficiency, driver approval, and customer service. To that end, we propose a thorough method to modeling, simulating, and optimizing the life of a delivery driver. The study's findings give delivery organizations important information about how to streamline their processes and enhance the overall satisfaction of both clients and drivers. This study advances the logistics and delivery sector and lays the groundwork for further research in the area by utilizing cutting-edge modeling methodologies and optimization strategies. In the end, we want our labor to have a good effect.**

***Index Terms*—delivery drivers, optimizing, difficulties.**

I. Introduction

Due to the phenomenal growth of e-commerce and the expanding popularity of online food delivery services, people's shopping and eating habits have altered. Delivery drivers now play a significant role in ensuring the fast and efficient delivery of goods and meals to customers as a result of the growth in demand. However, driving deliveries is not a simple job. It involves navigating congested streets, adhering to strict delivery schedules, and fighting against several other pressures that could harm their performance and sense of job satisfaction.

The study of modeling, simulation, and optimization of the life of a delivery driver becomes of utmost relevance in this situation. This study attempts to examine the complexity of a delivery driver's daily schedule, pinpoint problems, and offer creative ideas to improve operations and performance. It does this by using cutting-edge methods and tools.

Numerous obstacles that the delivery sector must overcome have an impact on its productivity and profitability. Route planning and optimization stand out among them as crucial areas with the potential to have a big impact on a driver's workload and delivery time. It is a difficult effort to balance the workload such that it is distributed fairly among the drivers and still complies with client delivery time constraints. Additional difficulties include driving through clogged traffic and running against unanticipated obstructions.

NCI, Modelling, Simulation and Optimisation, Dr Christian Horn

A. Research Question

What factors affect how long delivery drivers' daily routes are, and how may routing optimization techniques be used to shorten routes and improve delivery effectiveness?
Consumer Frequency and Diversity: The route's length can be considerably influenced by the consumers' geographic distribution. Drivers with clients grouped together may have shorter trips than those with consumers dispersed over a large area.

Traffic Condition: Poor road and traffic conditions can result in lengthier route and travel times. Routes chosen during periods of heavy traffic might be less effective.

Delivery Time Windows: Customer delivery time windows can have an impact on route planning in terms of time limitations. To meet particular delivery deadlines, drivers might have to travel longer routes.

B. Data Description

In result, it is likely that this function call generates and processes information about a network or graph that has 30 nodes and 100 clients. If given, the function can also plot the data and use logarithmic scaling. However, we would need to look at the generateData function's implementation and comprehend the precise issue it is meant to solve in order to properly comprehend the functionality and intent of this method.

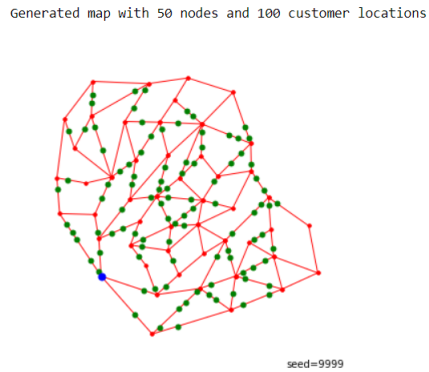

Fig. 1. Real Sample Data

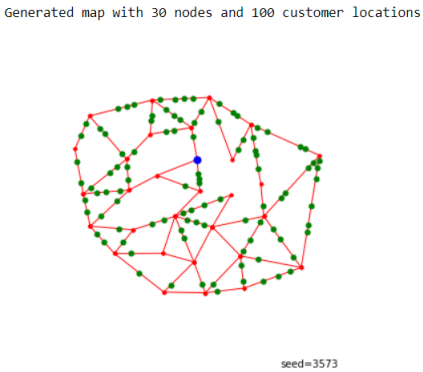

Fig. 2. My Data

C. Statistical Analysis

A crucial part of data-driven research and decision-making is statistical analysis. It involves examining, interpreting, and drawing conclusions from data using statistical techniques. Statistical analysis can be used in the context of delivery drivers' daily routes and route optimization to get insights into numerous aspects of the delivery process.

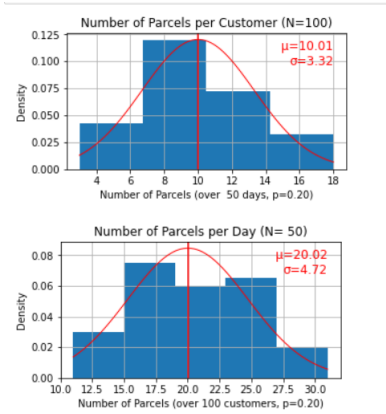

Fig. 3. Statistical Analysis p=0.2

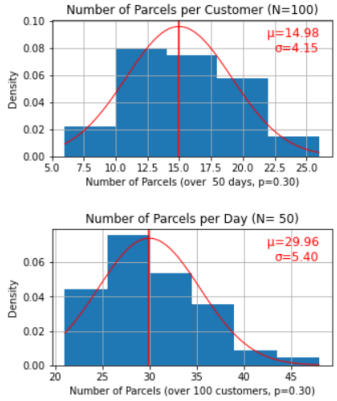

Fig. 4. Statistical Analysis p=0.3

A precise model to determine the Ideal Delivery route was created by the investigation discussed in this paper. The structure of this essay is as follows. The literature review of the Delivery Route System and its components is presented in Section II. The experimental findings are shown and discussed in Section III. The work is concluded in Section IV, which also offers further recommendations for enhancing the classification in subsequent studies.

II. Literature Review

Both simulation researchers and practitioners have given simulation optimizing a lot of attention. In this section, we give a general overview of simulation optimization as well as a number of strategies that have been put forth to address simulation optimization issues. Both continuous and discrete issues are covered, however the focus is on discrete issues and workable solutions to such issues. Olafsson & Kim (2002)
In spite of sampling some recent algorithmic and theoretical advancements in simulated optimisation research, we present an in-depth examination of the primary methods for performing simulator optimization. Then, review some of the software that is available for workbooks and simulation programs and offer a number of interesting applications. Fu et al. (2005)
To determine the possible effects of using cargo bikes and crowdsourcing in package delivery, a study was conducted in the medium-sized Italian city of Turin. The numerical studies demonstrate that integrating crowd drivers and green carriers into standard van delivery is advantageous due to cost- and resource-saving measures, while delivery effectiveness is reduced. In our studied cases, there is a reduction in CO2 emissions and overall travel distance. In addition, while handling the numerous internet inquiries, green carriers and crowdsourcing are promising and adaptable options. Perboli et al. (2022)
The Moodle Project, which was offered in the S3/S4 segment, served as the main source of inspiration for this study.

III. Methodology

The steps below can be used to execute for all data :

A. Finding All Shortest Paths

In a weighted, directed graph, the Floyd-Warshall algorithm is a dynamic-programming-based method for determining the quickest paths among all pairs of vertices. Even when has negative-weight edges (but not negative cycles), it successfully solves the (APSP) problem.

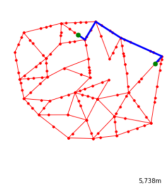

Fig. 5. Route 1

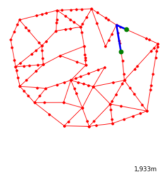

Fig. 6. Route 2

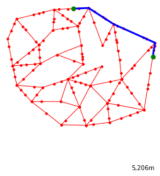

Fig. 7. Route 3

B. Shortest Path Finding- A* Algorithm

To determine the shortest route between a beginning node and a goal node in graphs and grids, the A* (pronounced "A-star") algorithm is a well-liked and effective pathfinding technique. Robotics, video games, and other fields that require determining the best pathways frequently employ it. The A* method finds the shortest path and is complete (it ensures finding a solution if one exists).

C. Model

- Processing
- In transit (from manufacture to distribution centre)
- Arrived in distribution centre
- Ready for delivery
- Out for delivery
- Customer not present
- Returned to distribution centre
- Delivered

The Parcel class was created to replicate and log events that occur during a parcel's transit through a delivery system. The class keeps track of the parcel's owner, destination, status, and time via attributes. While private methods take care of the internal time and event recording, public methods enable interaction with the parcel from outside parties and allow them to update its status. The packages manufactured throughout the simulation are all tracked by the REGISTER class variable.

In a mock parcel delivery system, customers are represented by the Customer class. Customer characteristics including location, response probability (q), and packages received are tracked. A factory function is also included in the class to guarantee that there is only one customer object per distinct location. When a package is delivered, the responseTime method determines the customer's reaction time, whereas the acceptParcel method documents the customer's acceptance of the package and logs the event.

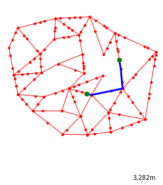

Fig. 8. Route 4

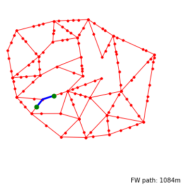

Fig. 9. Route 1

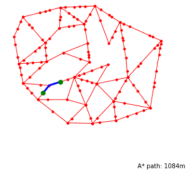

Fig. 10. Route 2

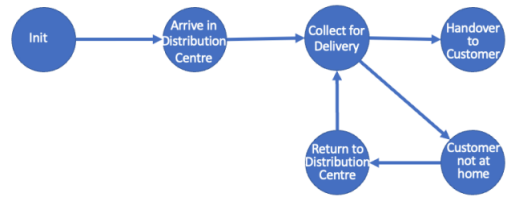

Fig. 11. Workflow

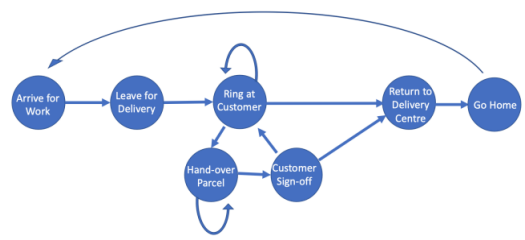

Fig. 12. Delivery system

Fig. 13. Delivery Centre

Carries out a more thorough simulation of a delivery driver's job routine, including customer encounters, delivery timings, and customer reactions. The simulation seems to be created for a discrete-event simulation framework, in which events are planned to take place at precise times to enable simulation progression. The implementation of the missing elements and their interaction with the Driver class determine the efficacy and validity of the code.

The code implements a simulation function with the name simulation, simulating the delivery of packages to a specific set of clients over a predetermined number of days in a hypothetical delivery system. The function accepts a number of parameters to specify the simulation parameters and returns a Recorder object that contains the data that was captured throughout the simulation.

D. Model Verification

Run Small Simulation (4 days)

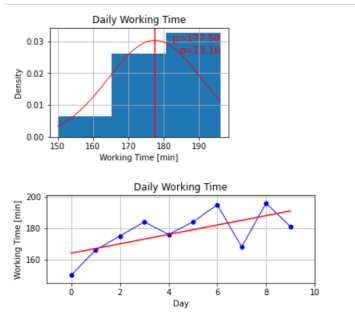

Fig. 14. Daily Working time

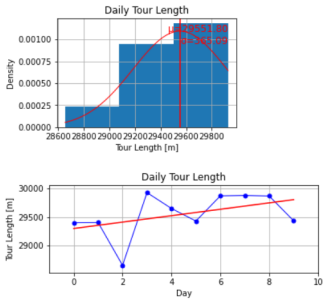

Fig. 15. Daily Tour time

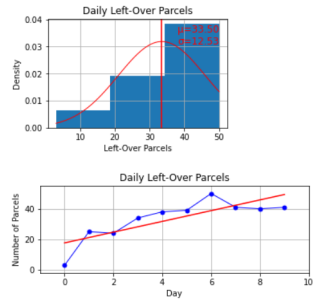

Fig. 16. Daily Left-over Parcels

E. Test Model I

Simulation days=50, p=0.2, limit=30000

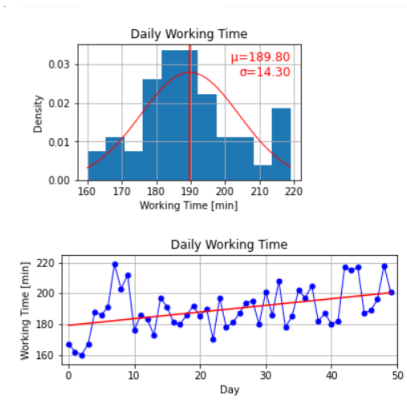

Fig. 17. Daily Working time

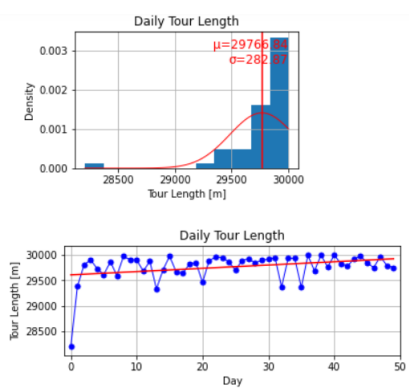

Fig. 18. Daily Tour time

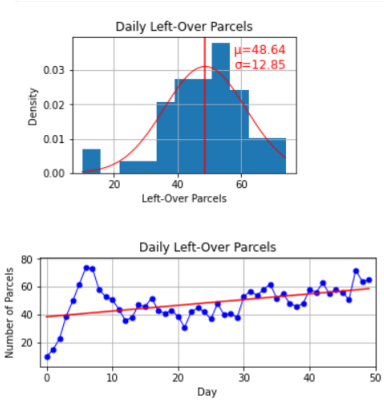

Fig. 19. Daily Left-over Parcels

F. Test Model II

Simulation days=50, p=0.3, limit=40000

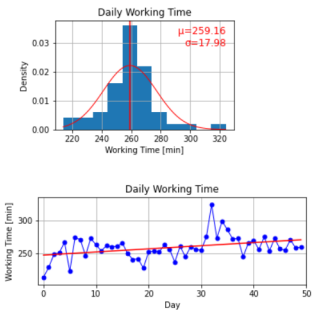

Fig. 20. Daily Working time

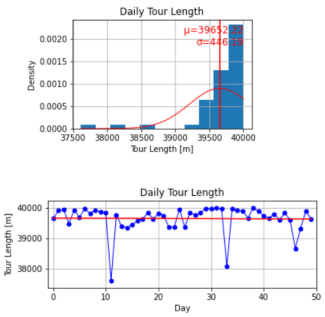

Fig. 21. Daily Tour time

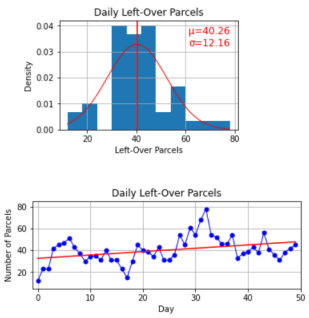

Fig. 22. Daily Left-over Parcels

G. Test Model III

Simulation days=50, p=0.21, limit=30000

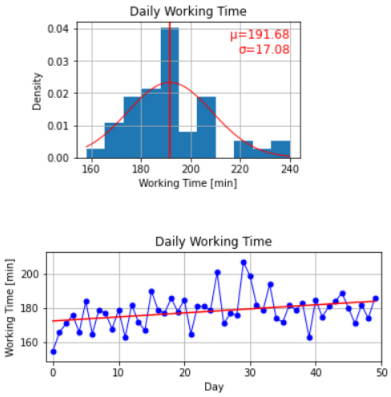

Fig. 23. Daily Working time

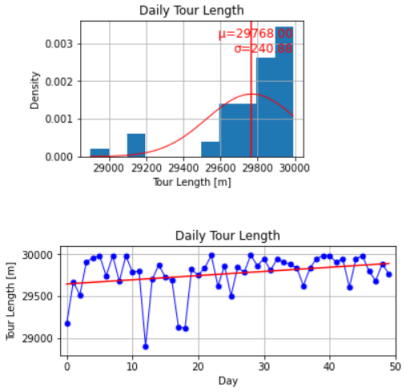

Fig. 24. Daily Tour time

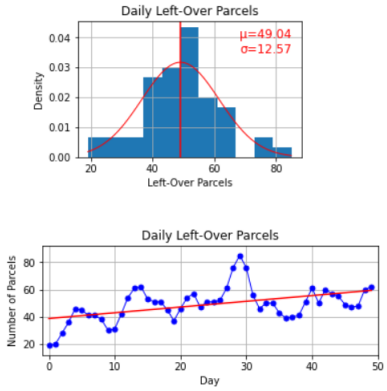

Fig. 25. Daily Left-over Parcels

H. Test Model IV

Simulation days=50, p=0.3, limit=30000

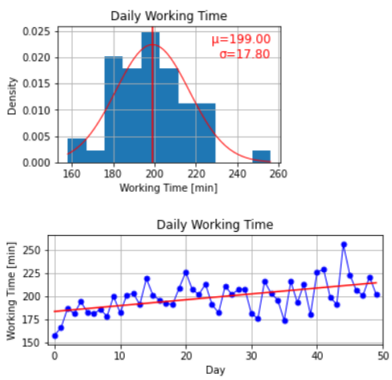

Fig. 26. Daily Working time

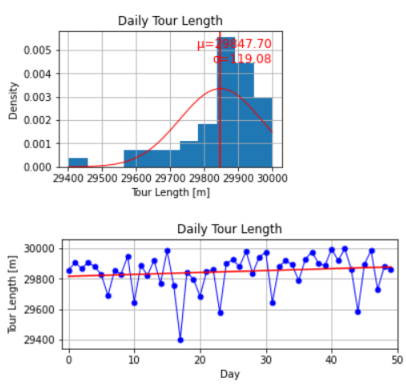

Fig. 27. Daily Tour time

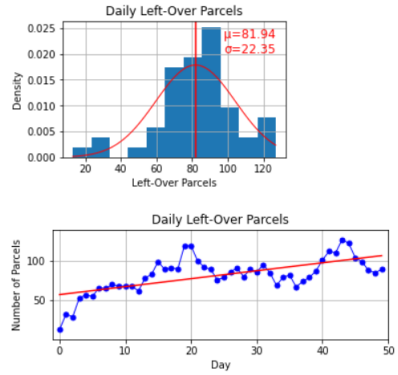

Fig. 28. Daily Left-over Parcels

I. Test Model V

Simulation days=50, p=0.2, limit=30000

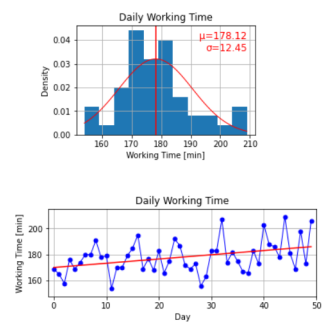

Fig. 29. Daily Working time

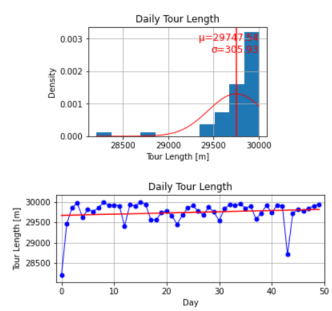

Fig. 30. Daily Tour time

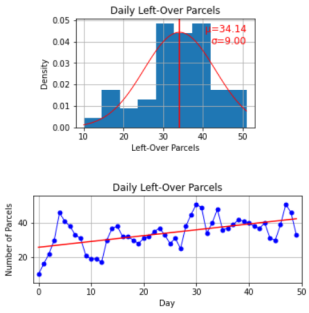

Fig. 31. Daily Left-over Parcels

IV. Results and Interpretation

None:	227	23.0%	None:	509	34.0%	None:	218	20.8%	None:	292	19.5%	None:	59	30.6%
1 days:	203	20.6%	1 days:	576	38.5%	1 days:	269	25.6%	1 days:	307	20.5%	1 days:	72	37.3%
2 days:	219	22.2%	2 days:	175	11.7%	2 days:	249	23.7%	2 days:	289	19.3%	2 days:	30	15.5%
3 days:	116	11.8%	3 days:	125	8.3%	3 days:	103	9.8%	3 days:	246	16.4%	3 days:	18	9.3%

Fig. 32. Over-View

In summary, Model V has less leftover parcel than Models I, II, III, and IV. Therefore, of the I, II, III, and IV models, the V model is the best and requires the least amount of work. Additionally, we can see increased precision in the delivery system through simulation analysis, and model V performed best when compared to models I, II, and III, while model IV performed worst.

Additionally, the ideal stimulation occurs when p=0.2, Distances = 30km, and Days = 50. Other than these findings, as the study reveals, they are not good.

The A* Algorithm, Floyd-Warshall is used in research algorithm to discover the shortest path along the specified route. between figures 5 to 10, respectively. It is offered an analysis to determine the shortest route.

V. Future Work

The modeling, simulation, and optimization of a delivery driver's life can advance to produce a more effective, long-lasting, and driver-centric delivery ecosystem by concentrating on these upcoming study areas. These developments may result in higher customer service standards, lower operational costs for logistics providers, and a favorable effect on the delivery sector as a whole. Using a variety of combinations as P and R variables will allow us to better optimize our results.

References

Fu, M., Glover, F. & April, J. (2005), 'Simulation optimization: a review, new developments, and applications', pp. 13 pp.–.

Olafsson, S. & Kim, J. (2002), 'Simulation optimization', **1**, 79–84 vol.1.

Perboli, G., Rosano, M. & Wei, Q. (2022), 'A simulation-optimization approach for the management of the on-demand parcel delivery in sharing economy', *IEEE Transactions on Intelligent Transportation Systems* **23**(8), 10570–10582.